Type MS · Recall the elements and building blocks of food molecules

Answer 1 of 7 · Question 3 part a.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
3 (a)	(i)	A (amino acids); B and D are components of lipids C is a carbohydrate		1
	(ii)	• add Biuret solution to test solution; • mauve/lilac, protein present; • remains blue, protein absent;		3
(b)	(i)	• suitable axes labels; • suitable scale; • correct plots; • suitable curves; • curves labelled;	max 3 for bar chart accept key	5
	(ii)	• denaturation; • loss of shape of active site/enzyme; • substrate no longer fits/decreases rate of activity;		3
	(iii)	any three from • B is much more active/works faster; • over whole temperature range; • higher optimum temperature for B/works fastest at 80°C; • active over a greater range of temperatures;		3

Total 15 marks

Type MS · Recall the elements and building blocks of food molecules

Answer 1 of 7 · Question 3 part a.i · 1 mark

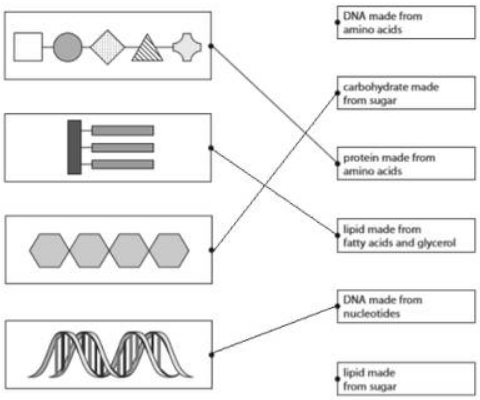
2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Recall the elements and building blocks of food molecules

Answer 2 of 7 · Question 3 part a · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

3	(a)	( ) 	Do not allow more than one line from each structure	3
	(b)	**D** stomach;		1
	(c) (i)	3/three		1
	(ii)	not washing equipment/transfer of food containing protein into distilled water;	Allow any valid alternative	1
	(d)	Hazard: Broken glass - cuts; Reducing risk: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables;	Ignore wear gloves	1
		Hazard: Biuret reagent - irritant; Reducing risk: Wear goggles/wash hands after use/don't use if skin is sensitive;	Ignore wear gloves	1

Total question 3 = 8 marks

Type MS · Recall the elements and building blocks of food molecules

Answer 2 of 7 · Question 3 part a · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Recall the elements and building blocks of food molecules

Answer 3 of 7 · Question 2 part a.ii · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a) (i)	C; (lipase)		1
(ii)	A; (carbon, hydrogen, oxygen)		1
(iii)	C; (liver)		1
(b)	• show effect of enzyme (1) • show that bile salts on their own don't break down fat (1)	Ignore control	2
(c)	• create alkaline/optimum pH (1) • for enzyme to work quickly (1)	Allow increases/adjusts pH	2
(d)	• increase rate of digestion (1) • with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1) • occurred after 5 minutes with bile salts present/digested more quickly in tube B (1)		3

Total 10 marks

Type MS · Recall the elements and building blocks of food molecules

Answer 3 of 7 · Question 2 part a.ii · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Recall the elements and building blocks of food molecules

Answer 4 of 7 · Question 7 part a.i · 1 mark

2022 JAN · Paper 01 · find part a.i in this mark scheme extract

Question number					
7	(a)	(i)	carbon hydrogen and oxygen;		1
		(ii)	nitrogen	Allow sulfur	1
		(iii)	19.6 - 12; 7.6 g	Allow ECF	2
		(iv)	protein 1.2 g × 17 = 20.4 kJ; lipid 4.8 g × 37 = 177.6 kJ; total = 198 kJ;	Allow ECF for 2nd and 3rd mark	3
	(b)		add alcohol to fish; add water; cloudy/milky appearance/emulsion indicates lipid;		3

Total 10 marks

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Type MS · Recall the elements and building blocks of food molecules

Answer 5 of 7 · Question 7 part a.ii · 1 mark

2022 JAN · Paper 01 · find part a.ii in this mark scheme extract

Question number					
7	(a)	(i)	carbon hydrogen and oxygen;		1
		(ii)	nitrogen	Allow sulfur	1
		(iii)	19.6 - 12; 7.6 g	Allow ECF	2
		(iv)	protein 1.2 g × 17 = 20.4 kJ; lipid 4.8 g × 37 = 177.6 kJ; total = 198 kJ;	Allow ECF for 2nd and 3rd mark	3
	(b)		add alcohol to fish; add water; cloudy/milky appearance/emulsion indicates lipid;		3

Total 10 marks

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Type MS · Recall the elements and building blocks of food molecules

Answer 6 of 7 · Question 5 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

5 (a)	• two monosaccharides; • joined together;		2
(b)	molecule too large;		1
(c) (i)	• transcription (of lactase gene) does not occur; • RNA is not produced/synthesised; • lactase/protein/enzyme not produced;		3
(ii)	• lactose can't be digested; • into components to be absorbed;		2
(d)	• they eat more dairy produce; • being able to digest it would give them an advantage;		2
(e)	• water passes from cells; • into gut; • diarrhoea causes poor absorption of vitamin D and Ca^{2+}; • as they are water soluble;		4

Total 14 marks

Type MS · Recall the elements and building blocks of food molecules

Answer 6 of 7 · Question 5 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Recall the elements and building blocks of food molecules

Answer 7 of 7 · Question 4 part a.i · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

4 (a)	(i)	• glucose (units); • linked/joined together;		2
	(ii)	insulin;		1
	(iii)	glucagon:		1
(b)	(i)	• $67.0 \div 1.69^2$; • 23.5/23.4/23.46/23.458;		2
	(ii)	healthy weight *less than 18.5 underweight* *18.5 to 24.9 healthy weight* *25 to 29.9 overweight* *30 to 34.9 obese* *35 to 39.9 severely obese* *more than 39.9 morbidly obese*	ECF value in (b)(i)	1
(c)		Any two from: • fatty deposits/plaques/atheroma; • in coronary arteries; • reduced blood flow to heart muscles/tissue; • lack of oxygen/glucose to heart muscle/tissue;		2

Total Marks 9

Type MS · Recall the elements and building blocks of food molecules

Answer 7 of 7 · Question 4 part a.i · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks